

In22-S5-CS3501 Data Science and Engineering Project
Department of Computer Science and Engineering
University of Moratuwa

SinhaSpeech

Sinhala Speech-to-Text AST Model

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Project ID: P13

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1. Executive Summary

Sinhala is still a low-resource language for speech technology. A handful of Whisper fine-tunes exist for it, but they are general-purpose and not accurate enough to build much on top of. This project is split into two phases.

Phase 1 : which is the core and guaranteed scope of the project, fine-tunes the base multilingual Whisper model directly (rather than an already Sinhala-tuned checkpoint) on collected Sinhala corpora, with the aim of building a stronger general-purpose Sinhala ASR model and pushing WER below what current fine-tunes achieve.

Phase 2: If Phase 1 succeeds and there is time left in the project timeline, Phase 2 takes the resulting model and wraps it in an accessibility application for two groups of students that existing Sinhala ASR tools do not currently serve: Deaf and hard-of-hearing (DHH) students, who need a teacher's spoken lecture converted into readable captions in real time, and students with writing difficulties, who need their spoken answers converted into editable text.

2. Problem Statement

Two groups of students in Sri Lankan classrooms are consistently underserved by the speech technology that currently exists for Sinhala:

- Deaf and hard-of-hearing (DHH) students, who do not need their own speech transcribed, but do need the teacher's spoken lecture turned into readable text in real time, or close to it, so they can follow along with the lesson.
- Students with writing difficulties, for example dysgraphia, motor impairments, or anything else that makes handwriting or typing slow and effortful, could complete classwork and answer questions far more easily if they could speak the content instead of writing it.

Underneath both of these use cases sits a more basic problem: Sinhala ASR itself is not yet accurate enough. The Whisper fine-tunes that are currently available are general-purpose (news, casual dictation) rather than built for either use case, and even on their own terms they are not especially accurate which is high enough to make any downstream application built on top of them unreliable. Before either student group can realistically be served, we first need to establish whether a Sinhala ASR model can be pushed to a level of accuracy that is actually usable in a classroom setting. That is the central focus of this project; the accessibility application is the applied outcome if we get there.

3. Data Description

We utilize a multi-source approach to ensure both broad coverage and high-quality benchmarking.

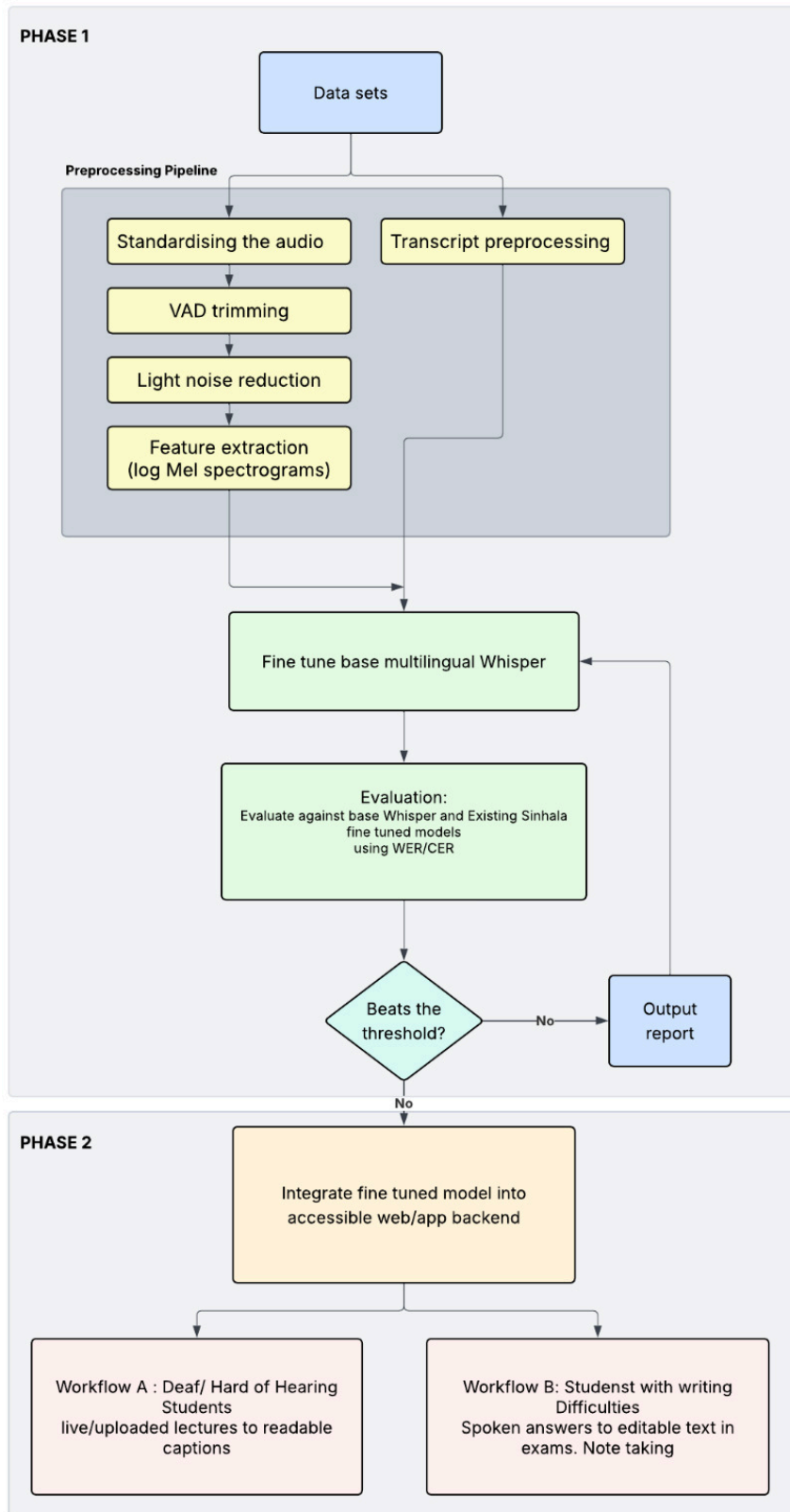
- **OpenSLR52 (Sinhala ASR corpus):** our primary training corpus. It's the same base corpus most existing Sinhala Whisper fine-tunes use, giving us a fair basis for comparison, and it's large enough on its own to meaningfully adapt Whisper to Sinhala. It consists of around 224 hours of transcribed speech, from 478 speakers, collected via crowdsourcing in Sri Lanka (Google/SLR project, Kjartansson et al.)
Source: openslr.org/52
- **Mozilla Common Voice (Sinhala subset):** held out purely for evaluation, not training. Since it's collected differently (different speakers, recording conditions, sentence styles), it gives a more honest check on whether the model generalises to Sinhala speech rather than just memorising OpenSLR52's recording style
It consists of a collection of spontaneous responses to questions in Sinhala
Source: <https://mozilladatacollective.com/datasets/>
- **YouTube-Sinhala-ASR (~4k utterances):** A diverse, multi-domain dataset scraped from various Sinhala YouTube channels, including news, education, health, and technology. This set includes rich metadata (domain, speaker gender, noise levels, and code-mixing indicators), allowing us to analyze model performance across different contexts.
Source: <https://huggingface.co/datasets/SPEAK-ASR/youtube-sinhala-asr>
- **Biz-Brains-Academy-Sinhala (~979 rows):** A specialized subset focused on business-education content. This data is particularly valuable for training as well as testing the model's ability to handle code-mixed speech, where Sinhala and English are used interchangeably, a common feature in professional and academic settings in Sri Lanka.
Source: <https://huggingface.co/datasets/SPEAK-ASR/biz-brains-academy-sinhala>

The datasets are structured audio-transcript pairs, already segmented into short, utterance-level clips, so no manual segmentation or forced alignment is needed.

4. Methods

The project is split into two phases. Phase 1 is the core, guaranteed scope: fine-tuning and rigorously evaluating a Sinhala ASR model against existing baselines. Phase 2 is an extension we take on only if Phase 1 produces a genuinely competitive model and there is still time left in the project timeline.

Figure 1 gives the system-level view of how the two phases fit together.



4.1 Data Preprocessing (Phase 1)

Selected Data sets need a few preprocessing steps before they are consistent and usable for fine-tuning:

Step 1 - Standardising the audio. Every recording is converted to 16kHz, mono, WAV, since that is the format Whisper's feature extractor expects, and the two datasets will not necessarily match each other by default (different original sample rates, encodings, etc.).

Step 2 - VAD trimming. We run a pretrained Silero VAD model over each clip to trim leading and trailing silence. Whisper is known to hallucinate text when fed long stretches of silence, so trimming dead air at the edges before training helps avoid learning that habit.

Step 3 - Light noise reduction. We apply spectral gating (via the noisereduce library, non-stationary mode) as a light cleanup pass, removing low-level mic hiss or hum without needing a separate silence-only reference clip.

Step 4 - Feature extraction (log-Mel spectrograms). Once the audio is cleaned, it is converted into the log-Mel spectrogram format Whisper's encoder expects, using Whisper's own feature extractor from the Hugging Face transformers library (frame windowing, FFT, Mel filter bank, then log). We are not implementing this by hand, but it is worth understanding since it is the actual representation the model learns from.

Transcript preprocessing involves basic text normalisation - fixing obvious spelling inconsistencies, stripping stray symbols, and making sure Sinhala Unicode is represented consistently across both datasets. Sinhala Unicode can vary depending on the input method originally used to type it, so this step matters for keeping transcripts consistent for training.

4.2 Model Fine-Tuning (Phase 1)

In this phase, we will fine-tune the multilingual Whisper model for Sinhala speech recognition. We plan to begin with Whisper-Small as it provides a good balance between performance and computational requirements. If hardware limitations become significant, Whisper-Base will be used instead.

The model will be fine-tuned in a single stage using selected Sinhala speech datasets. The Mozilla Common Voice Sinhala dataset will not be used during training and will instead be reserved as an independent test set to evaluate the model's ability to generalize to unseen data.

The implementation will use the Hugging Face Transformers and Datasets libraries. Audio samples will be converted into log-Mel spectrograms, which serve as the model input, while the corresponding Sinhala transcripts will be tokenized as the target output for supervised learning.

Training will be carried out using mixed-precision (FP16) on a GPU-enabled environment such as Google Colab Pro or another cloud-based GPU platform to reduce memory usage and training time. To improve the model's robustness and reduce overfitting, SpecAugment (time and frequency masking) will be applied during training.

After each training epoch, the model will be evaluated on a held-out validation split of the OpenSLR52 dataset using Word Error Rate (WER) and Character Error Rate (CER). The checkpoint with the best validation performance will be selected as the final model instead of simply using the last training checkpoint.

4.3 Model Integration (Phase 2)

If the project proceeds to Phase 2, the fine-tuned Whisper model will be integrated into the proposed accessibility application as its speech recognition engine. No further training will be performed during this phase; the model checkpoint produced at the end of Phase 1 is treated as final, and the work shifts entirely to deploying it within a usable application for the two target student groups.

Practically, this means wrapping the fine-tuned model behind an inference API that the application's front end can call, and building the two workflows around it: real-time or uploaded lecture transcription for Deaf and hard-of-hearing students, and spoken-response transcription for students with writing difficulties. Supporting features such as transcript editing, export, audio upload/recording, and confidence-based highlighting of likely errors are also built at this stage, since they depend on having a working model to plug into.

5. Evaluation Plan

5.1 Phase 1 Evaluation

The performance of the fine-tuned model will be evaluated throughout the training process using both training and testing metrics.

During training, training loss, validation loss, learning rate, Word Error Rate (WER), and Character Error Rate (CER) will be logged and visualized using Weights & Biases (W&B). This enables us to monitor the training progress, compare different fine-tuning experiments, and identify overfitting at an early stage.

To evaluate the effectiveness of the Voice Activity Detection (VAD) preprocessing step, a small manually labelled audio subset will be used to measure VAD precision/recall verifying speech segments are correctly separated from silence without removing spoken content.

The final model will be evaluated on a held-out test set using WER and CER. Its performance will be compared against two baselines: the original multilingual Whisper model without Sinhala fine-tuning

and an existing published Sinhala Whisper fine-tuned model. This comparison will determine whether our fine-tuning approach provides an improvement over existing Sinhala ASR performance.

To assess the model's ability to generalize, the same evaluation will be carried out on the Mozilla Common Voice Sinhala dataset, which is not used during training. This provides an independent measure of performance on speech from a different source.

In addition to reporting WER and CER, a confusion-pattern breakdown will be done which represent phonemes or words the model most often mixes up, reported alongside the raw WER/CER numbers, since the error rate alone does not explain why the model is getting things wrong.

5.2 Phase 2 Evaluation

The application's response time will be measured by recording the time taken to generate a transcript from both uploaded audio files and live speech, ensuring that the system is suitable for near real-time captioning.

The reliability of the model's confidence scores will also be examined to determine whether higher confidence values correspond to more accurate transcriptions. This is important if confidence values are used to highlight potentially incorrect captions.

Finally, a small usability evaluation will be conducted with a limited number of users, or suitable representatives if direct access to the target users is not possible. Feedback will be collected on transcription accuracy, caption readability, ease of correcting errors, and the overall usability of the application. This feedback will be used to identify areas for further improvement.

6. Expected Outcomes and Success Criteria:

Expected Outcomes

A Sinhala Whisper fine tune trained from the ground up from public Sinhala dataset(s), with its training configuration documented.

A reproducible experimental methodology

A same test set benchmark against basic Whisper model and other Sinhala fine tunes.

An error analysis identifying the specific challenges remaining in Sinhala ASR.

Success Criteria

Measurably lower WER than base Whisper, and lower or comparable WER to existing public Sinhala fine tunes, on our held out test set.

At least three distinct fine tuning configurations completed and documented with comparable results. Error analysis surfaces concrete, specific failure patterns rather than aggregate numbers alone.

7. Division of work (individual responsibilities)

Yohan Jayasinghe (230283A):

- Build and run the audio preprocessing pipeline (standardisation, VAD trimming, light noise reduction, feature extraction) across all selected datasets.
- Run Fine-Tuning Configuration 1 (Whisper-small on OpenSLR52 with baseline hyperparameters) and log training/validation metrics to Weights & Biases.

Tharupahan Jayawardana (230295L):

- Set up dataset loading and observe the data set for training and testing.
- Run Fine-Tuning Configuration 2 (Whisper-small on OpenSLR52 with an adjusted learning-rate schedule and SpecAugment settings) and log results to Weights & Biases.
- Contribute to Phase 2 backend and model-serving integration.

Imindu Jayasekara (230277J):

- Run Fine-Tuning Configuration 3 (Whisper-base fallback configuration, or an alternative training-data mix) and log results to Weights & Biases.
- Build the evaluation pipeline - WER/CER computation, baseline comparisons, and confusion-pattern analysis and lead Phase 2 usability testing.

8. Preliminary Bibliography:

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